Sandoval County Water Data Report

Nov 12, 2026 · @James

Summary

The cleaning script turns 8 raw CSVs into 6 cleaned files. Together they cover Sandoval County groundwater levels, aquifer tests, well drilling logs, one stream gage, and aquifer sensitivity. The three stream-gage files are combined into one.

- **Groundwater is fairly shallow and mostly stable.** NMBGMR wells have a median depth to water of 57 ft. Four of the five wells measured in both 1997–98 and 2018–19 changed by less than 3 ft or rose.
- **Aquifers vary widely.** The median transmissivity is about 690 ft²/day in the Santa Fe Group, compared with 55 ft²/day in the Abo Sandstone.
- **The Rio Puerco at Cuba is often dry.** It had no flow at 6 of 10 visits from 2019 to 2026. All flowing visits were in April or May, at 7–20 cfs.
- **The sensitivity layer is statewide.** Its polygons add up to about the area of New Mexico, and 47.5% of that area is rated High.
- **The cleaning flagged 15 data-quality problems.** Most were fixed. A few still need a decision (see Data quality).

Datasets

Each cleaned file starts with the same columns: `site_id, site_name, latitude, longitude, date, source`. The exception is the sensitivity file, which has map polygons instead of sites.

Cleaned file	Raw input	Source	Rows	Sites	Years
groundwater_levels_monthly.csv	water_flow_data_sandoval_county_sources.csv	USGS	2,673	9 wells	not reviewed
groundwater_levels_nmbgmr.csv	data2_groundwater_sandoval_county.csv	NMBGMR	451	87 wells	1997–2019
aquifer_tests.csv	ose_aquifer_test_wells.csv	NM OSE	179	73 wells	1930–2008
streamflow_measurements.csv	data4 channel, field and metadata files (3)	USGS	10	1 gage	2019–2026
well_logs.csv	data7_Wells With Well Log Information.csv	NM OSE	6	6 wells	1979–2022
aquifer_sensitivity.csv	data3_Aquifer_Sensitivity.csv	Sensitivity map layer	613 polygons	none	no dates

The USGS monthly row counts come from your MATLAB run. That raw file wasn't shared with me, so this report doesn't analyze it further.

Groundwater levels

In the NMBGMR data, the median depth to water is 57.5 ft. Half of all readings fall between 35.6 and 119.9 ft. The shallowest reading is 1.7 ft (PB-0005) and the deepest is 399.3 ft (PC-084).

Most of the record comes from one monitoring program. 383 of the 451 readings were taken in 1997–98, 18 in 1999, and 50 in 2016–2019. 48 wells have 5 or more readings, and 32 have only one.

Well ID prefix	Wells	Readings	Median depth to water (ft)
PC	48	325	67.6
NM	13	88	45.4
PB	24	32	21.1
WL	2	6	67.1

Wells measured in both 1997–98 and 2018–19:

Well	1997–98 (ft)	2018–19 (ft)	Change
PC-002	139.5	139.5–141.2	Stable
PC-019	167.4–170.2	167.6	Stable
PC-055	204.8–208.4	210.9	About 2.5 ft deeper
PC-014	56.6–59.2	108.7	About 50 ft deeper (one reading; check it)
PC-025	138.9 falling to 58.6	30.6	Rose about 108 ft

PC-025 was completed in October 1996, and its level rose steadily through 1997–98. This looks like the well recovering after drilling, not a change in the aquifer. PC-014's 2019 reading is one point, so it should be confirmed before anyone treats it as a decline.

226 of the 451 readings have a level-status code. Most are R (recently pumped, 177) and T (nearby well recently pumped, 34). These readings may not be static water levels, so filter on `level_status` before analyzing trends.

The USGS monthly file adds 2,673 monthly-mean levels at 9 wells. It uses the same first columns, so it can be stacked with the NMBGMR data.

Aquifer tests

There are 179 tests at 73 wells, run from 1930 to 2008 (median year 1994). Across the 117 tests that report it, the median transmissivity is 100 ft²/day. It differs by more than 10 times between formations.

Formation	Tests	Tests with transmissivity	Median transmissivity (ft²/day)
Santa Fe Group	27	14	690
Madera Limestone	14	13	668
Agua Zarca Sandstone Member	4	4	154
Abiquiu Tuff	7	7	88
Ojo Alamo Sandstone	8	5	58
Abo Sandstone	31	29	55

54 tests have no formation listed. Five of the seven Abiquiu Tuff tests have negative pumping rates, so treat that formation's median with caution.

Other medians, with the number of tests that report each value:

- Well depth: 405 ft (137 tests), ranging from 30 to 6,356 ft
- Starting water level: 189 ft (125 tests)
- Pumping rate: 15 gpm (139 tests)
- Specific capacity: 0.6 gpm/ft (89 tests)
- Hydraulic conductivity: 2.7 ft/day (50 tests)
- Storativity: 0.0005 (only 14 tests)

104 water-level dates were the 1/1/1900 "unknown" placeholder, and the script blanked them. 30 test dates fall on January 1, so some may only be accurate to the year.

Streamflow: Rio Puerco at Cuba

The USGS gage 08332525 (drainage area 107.2 sq mi) had no flow at 6 of 10 visits between April 2019 and March 2026. Those dry visits were in August, October, December and March. All four flowing visits were wading measurements in April or May.

Date	Flow (cfs)	Width (ft)	Area (ft²)	Velocity (ft/s)	Gage height (ft)	Bed	Rating
2024-04-30	20.2	11.0	9.67	2.09	3.05	Gravel	Fair
2021-04-14	6.89	11.5	7.00	0.98	3.04	Silt/mud	Poor
2020-05-04	20.4	12.0	13.3	1.53	4.61	Silt/mud	Poor
2019-04-25	19.4	12.5	13.6	1.42	2.95	Sand	Fair

In every flowing visit, flow matches area × velocity within 1%. Gage height doesn't predict flow well, though: 20.4 cfs read 4.61 ft in 2020, and 20.2 cfs read 3.05 ft in 2024. The bed changed from silt/mud to gravel between those visits, so the channel shifted.

Measurements 12–15 are missing from the file, so there is nothing between April 2021 and April 2024. Gage height comes from the field-measurements file and matches only the four flowing visits.

Aquifer sensitivity

High sensitivity covers 47.5% of the mapped area, even though it has fewer polygons than Moderate.

Class	Polygons	Share of area
High	152	47.5%
Moderate	339	36.0%
Less	122	16.5%

The areas are in square degrees. One square degree is about 10,000 km² at this latitude. All 613 polygons add up to about 31 sq deg, or roughly 310,000 km². That's about the area of New Mexico. The largest single polygon (High, 1.33 sq deg) is bigger than Sandoval County. So this is a statewide layer, not a Sandoval-only one.

The CSV has only attributes, with no polygon shapes. It can't be matched to wells until you get the shapefile or GeoJSON for this layer.

Well logs

The six wells are in the San Juan (SJ) sub-basin. Borehole depths run from 100 to 680 ft, and drillers first hit water between 114 and 480 ft down.

Well	Finished	Borehole depth (ft)	First water (ft)	Location from
SJ 04531 POD1	2022-06-30	600	340	GPS/survey
SJ 04455 POD1	2021-05-20	630	480	GPS/survey (bad northing)
SJ 04288 POD1	2018-07-18	407	114	GPS/survey
SJ 03562	2005-03-26	680	240	PLSS (approximate)
SJ 01824	1984-06-20	100	not recorded	PLSS (approximate)
SJ 00809	1979-04-19	322	145	PLSS (approximate)

SJ 01824 and SJ 03562 have identical coordinates. That's because both were placed at the center of the same quarter-quarter section, not because they're the same well. Where it was recorded, first water was 28–76% of the way down the borehole.

Data quality

The script fixed or flagged 15 problems. Three need your decision; they're marked **Open**.

Dataset	Problem	What the script does
Aquifer tests	Negative values in 7 tests: 5 pumping rates, 3 water levels (see the next section)	**Open.** Kept as is. A negative water level may mean a flowing (artesian) well, but a negative pumping rate is an error.
Groundwater (NMBGMR)	PC-014 is 50 ft deeper in 2019 than in 1998	**Open.** Kept as is. One reading, so confirm it before using it.
Well logs	License 1357 is listed as both "BAILEY, MARK" and "BAILEY, MARKMANLENENER"	**Open.** Flagged. Probably a typo for the same driller.
Well logs	SJ 04455 northing puts it about 80 km north, outside the county	Lat/lon blanked; raw UTM kept so it can be corrected
Well logs	SJ 01824's log was filed before drilling started	Flagged, kept
Well logs	Padded IDs and a footnote row at the bottom of the file	IDs trimmed, footnote row dropped
Groundwater (NMBGMR)	County spelled both "SANDOVAL" and "Sandoval"	Treated as the same
Groundwater (NMBGMR)	Coordinates given only in UTM	Converted to lat/lon
Groundwater (NMBGMR)	5 dry readings with no depth	Dropped
Groundwater (NMBGMR)	PB-0019 casing diameter of 41 (others are 0.3–0.6 ft)	Blanked
Groundwater (NMBGMR)	NM-26928's water is deeper than the well (103 ft vs 100 ft)	Flagged, kept
Groundwater (NMBGMR)	PC-020's time fields are swapped ("2:13 PM" in the time-zone column)	Time not exported
Aquifer tests	104 water-level dates are the 1/1/1900 placeholder	Blanked
Aquifer tests	A range ("11 to 21") where a single number belongs	Replaced with the midpoint, 16

Dataset	Problem	What the script does
Streamflow	"Unspecified" categories; "main" vs "Main"; measurements 12–15 missing	Blanked; capitalization fixed; gap reported

Negative values in the aquifer tests

All five negative pumping rates come from one 1980 test series at RG-24834, wells SH-1 to SH-5 in the Abiquiu Tuff. A pumping rate can't be below zero, so these were most likely entered with the wrong sign. The well comments say "7 wells all have same RG-24834".

OSE POD ID	Well	Test date	Formation	Well depth (ft)	Pumping rate (gpm)	Start water level (ft)	Transmissivity (ft²/day)
RG-24834	SH-1	1980-05-28	Abiquiu Tuff	450	**−5.22**	366.92	2.8
RG-24834	SH-5	1980-05-30	Abiquiu Tuff	450	**−5.05**	371.84	20.9
RG-24834	SH-2	1980-06-03	Abiquiu Tuff	425	**−14.15**	383.01	423.1
RG-24834	SH-3	1980-06-03	Abiquiu Tuff	450	**−13.38**	not recorded	88.1
RG-24834	SH-4	1980-06-04	Abiquiu Tuff	450	**−14.24**	379.74	34.2

The three negative water levels are probably not errors. A negative depth to water usually means the water stood above the ground, as in a flowing (artesian) well.

OSE POD ID	Well	Test date	Formation	Pumping rate (gpm)	Start water level (ft)	End water level (ft)
RG-51861	Wild Cherry Farm	2006-10-23	Abo Sandstone	51.4	**−25.84**	not recorded
RG-76797	Expl-3	2002-12-12	Agua Zarca Sandstone	10.4	**−9.1**	**−2.4**

The cleaning script currently keeps all of these values as they are. To correct the pumping rates, either flip their sign (if the original test report confirms the magnitudes) or blank them. Leave the negative water levels and note that they mean flowing wells.